Lecture 16

Energy and global existence

Lecture 15 constructs the nonlinear field for a short time. Its continuation theorem reduces global existence to a bound on the perturbation's energy-space norm. The conserved nonlinear energy supplies that bound on every finite time interval.

Objectives. *We prove energy conservation at the regularity of the constructed solution. We then exclude finite-time termination for perturbations of either vacuum and of the kink. A focusing example shows why the sign of the potential matters.*

By the end of this lecture you will be able to:

- Prove conservation by spatial smoothing of a weak energy solution.
- Bound the full perturbation norm on every bounded time interval.
- Use the continuation alternative and distinguish positivity from conservation.

16.1 Energy on an affine background

Retain $V(s) = \frac{1}{2}(1-s^2)^2$ and $b \in \{1, -1, \tanh x\}$ from Lecture 15. Write $\phi = b + v$ and $p = v_t$, where $v \in C(I; H^1(\mathbb{R})) \cap C^1(I; L^2(\mathbb{R}))$ on an interval I containing zero.

Definition 16.1 (Nonlinear energy). The *nonlinear energy* of $\phi = b + v$ with velocity p is

$$E_{\mathrm{en}}[\phi, p] = \frac{1}{2}\int_{\mathbb{R}} \left[p^2 + \phi_x^2 + (1-\phi^2)^2\right] \mathrm{d}x, \qquad \phi_x = b' + v_x. \tag{16.1}$$

Write $E_{\mathrm{en}}(t)$ for its value along the field.

This integral is finite and continuous in time. Indeed, $b' \in L^2$ and $1 - b^2 \in L^2$ for each allowed background. For the kink both are $\operatorname{sech}^2 x$, and for a vacuum both vanish. The one-dimensional embedding of Lemma 15.2 makes v bounded. The factorisation

$$1 - (b+v)^2 = (1-b^2) - 2bv - v^2$$

therefore belongs to L^2, continuously in time. No L^2 condition on b or ϕ is required. The same argument gives $V'(\phi) = 2\phi(\phi^2 - 1) \in C(I; L^2)$.

16.2 Conservation at energy regularity

Lemma 16.2 (Conservation for energy solutions). *Every energy solution of Lecture 15 satisfies $E_{\mathrm{en}}(t) = E_{\mathrm{en}}(s)$ for $s, t \in I$.*

Proof. We smooth the given solution only in space and estimate its nonlinear residual. Fix a compact time interval $K \subset I$. Choose $\vartheta \in C_c^\infty(\mathbb{R})$ nonnegative, with integral one, and set

$\vartheta_\varepsilon(x) = \varepsilon^{-1}\vartheta(x/\varepsilon)$. A subscript ε denotes convolution in x. In particular, $\phi_\varepsilon = b_\varepsilon + v_\varepsilon$ is well-defined because b is bounded.

First we justify convergence in the affine energy class. The background may fail to lie in L^2, but its translation difference obeys

$$\|b(\,\cdot - h) - b\|_2 \le |h|\|b'\|_2.$$

This follows from $b(x-h)-b(x) = -\int_0^h b'(x-s)\,\mathrm{d}s$ and the norm triangle inequality. Averaging over h gives

$$\|b_\varepsilon - b\|_2 \le \varepsilon\|b'\|_2 \int |y|\vartheta(y)\,\mathrm{d}y \longrightarrow 0, \qquad (b_\varepsilon - b)' = (b')_\varepsilon - b' \longrightarrow 0 \text{ in } L^2.$$

The last limit is the approximate-identity theorem from *Analysis* I, Lecture 5. Applying that theorem to v, v_x and p gives

$$\phi_\varepsilon - \phi \longrightarrow 0 \text{ in } C(K;H^1), \qquad p_\varepsilon \longrightarrow p \text{ in } C(K;L^2). \tag{16.2}$$

Uniformity in time follows from the compact-image, finite-net argument used in Lemma 15.4. Convolution with ϑ_ε is a contraction on L^2 and H^1, by the integral triangle inequality.

The local energy norm and the embedding bound ϕ uniformly on $K\times\mathbb{R}$. The nonnegative mollifier gives the same bound for ϕ_ε. If both absolute values are at most M, factorisation gives

$$\|(1-\phi_\varepsilon^2) - (1-\phi^2)\|_2 \le 2M\|\phi_\varepsilon - \phi\|_2.$$

Thus (16.2) proves uniform convergence of all three energy terms: $E_{\mathrm{en}}[\phi_\varepsilon, p_\varepsilon] \to E_{\mathrm{en}}$ on K.

Next, convolution of the distributional field equation gives

$$\partial_t^2\phi_\varepsilon - \partial_x^2\phi_\varepsilon + V'(\phi_\varepsilon) = R_\varepsilon, \qquad R_\varepsilon = V'(\phi_\varepsilon) - (V'(\phi))_\varepsilon. \tag{16.3}$$

This equation has enough time regularity to differentiate its energy. For fixed ε, its right side before adding $V'(\phi_\varepsilon)$ gives $\partial_t p_\varepsilon = \partial_x^2\phi_\varepsilon - (V'(\phi))_\varepsilon \in C(K;L^2)$. Here $\partial_x^2\phi_\varepsilon = \vartheta_\varepsilon' * (b' + v_x)$ is continuous in L^2. The norm-primitive argument of Lecture 4 turns this distributional equality into a norm derivative. Also $\partial_t(\partial_x\phi_\varepsilon) = \partial_x p_\varepsilon \in C(K;L^2)$. The embedding of Lecture 15 makes $\phi_\varepsilon = b_\varepsilon + v_\varepsilon$ continuous in L^∞, since b_ε is time-independent and $v_\varepsilon \in C(K;H^1)$. Its time difference quotients converge to p_ε in L^2. Factoring the difference of two squares therefore gives $\partial_t(1-\phi_\varepsilon^2) = -2\phi_\varepsilon p_\varepsilon$ in L^2. Consequently the three squared norms in the energy are differentiable.

Their spatial terms cancel by integration by parts. A cutoff equal to one on $[-R,R]$, with derivative bounded by C/R, leaves an error at most $(C/R)\|p_\varepsilon\|_2\|\partial_x\phi_\varepsilon\|_2$, which tends to zero. This uses the derivative of the field, not an L^2 norm of the background. The resulting energy identity is

$$\frac{\mathrm{d}}{\mathrm{d}t}E_{\mathrm{en}}[\phi_\varepsilon, p_\varepsilon] = \int_{\mathbb{R}} p_\varepsilon R_\varepsilon\,\mathrm{d}x. \tag{16.4}$$

Finally, V' is Lipschitz on the bounded range $[-M,M]$. The nonlinear residual satisfies

$$\|R_\varepsilon\|_2 \le C_M\|\phi_\varepsilon - \phi\|_2 + \|V'(\phi) - (V'(\phi))_\varepsilon\|_2 \longrightarrow 0 \quad \text{uniformly on } K.$$

The second term converges uniformly because $V'(\phi) \in C(K;L^2)$. The velocities p_ε are uniformly bounded in $C(K;L^2)$. Integrate (16.4) between any $s,t\in K$ and let $\varepsilon\downarrow 0$. The right side tends to zero, while the two endpoint energies converge to $E_{\mathrm{en}}(t)$ and $E_{\mathrm{en}}(s)$. This proves conservation on every such interval. ∎

16.3 The continuation norm stays finite

Theorem 16.3 (Global nonlinear evolution). *For $b \in \{1, -1, \tanh x\}$ and every $(v_0, g_0) \in H^1(\mathbb{R}) \times L^2(\mathbb{R})$, Lecture 15's solution is global in both time directions. Its nonlinear energy is the constant $E_0 = E_{\mathrm{en}}[b + v_0, g_0]$. For every $t \in \mathbb{R}$ it obeys*

$$\begin{aligned} \|v_t(t)\|_2 &\le \sqrt{2E_0}, \\ \|v(t)\|_2 &\le \|v_0\|_2 + |t|\sqrt{2E_0}, \\ \|v_x(t)\|_2 &\le \sqrt{2E_0} + \|b'\|_2. \end{aligned} \tag{16.5}$$

Proof. Prove the bounds first on the maximal local solution. Every term of (16.1) is nonnegative. Conservation therefore bounds $\|\phi_t\|_2$ and $\|\phi_x\|_2$ by $\sqrt{2E_0}$. The first equals $\|v_t\|_2$. The identity $v(t) = v_0 + \int_0^t v_t(s)\,ds$ holds in L^2, since v is C^1 there. Taking norms gives the middle bound, with reversed integration for $t < 0$. Finally, $v_x = \phi_x - b'$ gives the last bound.

These estimates bound all three terms of $\|(v, v_t)\|_{\mathrm{lin}}^2 = \|v\|_2^2 + \|v_x\|_2^2 + \|v_t\|_2^2$ on every bounded time interval. The blow-up alternative of Theorem 15.5 excludes a finite positive endpoint. Its time-reversed version excludes a finite negative endpoint. Uniqueness is the local uniqueness already proved there. ∎

Example 16.4 (A perturbation of the vacuum). Take $b = 1$, $v_0(x) = e^{-x^2}$ and $g_0 = 0$. Since $V(1 + v) = 2v^2 + 2v^3 + \frac{1}{2}v^4$, Gaussian integration gives

$$E_0 = \frac{5}{2}\sqrt{\frac{\pi}{2}} + 2\sqrt{\frac{\pi}{3}} + \frac{\sqrt{\pi}}{4}, \qquad \|v_0\|_2 = \left(\frac{\pi}{2}\right)^{1/4}.$$

The middle estimate in (16.5) becomes

$$\|v(t)\|_2 \le \left(\frac{\pi}{2}\right)^{1/4} + |t|\sqrt{2E_0}.$$

The spatial derivative and velocity are bounded by $\sqrt{2E_0}$ because $b' = 0$. Hence all parts of the existence norm stay finite on each finite time interval, even though the displayed L^2 estimate permits growth. For the kink, the same proof includes the fixed offset $\|b'\|_2$ in the spatial derivative bound.

These are continuation estimates. They allow the perturbation norm to grow with time. Lecture 18 needs further information before drawing a stability conclusion near a kink.

16.4 A focusing sign can conserve energy and still blow up

For this guard, change the spatial domain to $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and take the focusing equation

$$u_{tt} - u_{xx} = u^3, \qquad V_{\mathrm{foc}}(u) = -\frac{1}{4}u^4.$$

Fix $T > 0$. The spatially constant periodic field

$$u(t, x) = \frac{\sqrt{2}}{T - t}, \qquad 0 \le t < T, \qquad u_{tt} = \frac{2\sqrt{2}}{(T-t)^3} = u^3$$

is a classical solution. Its initial data are smooth and have finite periodic energy-space norm. Its energy is nevertheless zero at every time:

$$E_{\mathrm{foc}}(t) = 2\pi\left(\frac{1}{2}u_t^2 - \frac{1}{4}u^4\right) = 0, \qquad \int_{\mathbb{T}} (u^2 + u_x^2 + u_t^2)\,dx = 4\pi\left[(T-t)^{-2} + (T-t)^{-4}\right] \longrightarrow \infty.$$

The negative potential cancels the kinetic energy exactly. Conservation alone supplies no bound on this solution's existence norm. The domain is essential: these constant spatial data are finite-energy periodic data, and are not claimed to belong to $H^1(\mathbb{R})$.

Summary. For the one-dimensional ϕ^4 field near a vacuum or the kink, spatial mollification proves energy conservation for the actual weak energy solution. The nonlinear commutator tends to zero on every compact time interval. Positive energy controls the field's spatial derivative and velocity. Integrating the velocity controls the perturbation's L^2 norm for finite time. These bounds exclude the finite-lifespan alternative and give global existence. A focusing potential can lose precisely the needed norm control.

Self-check. Why is the nonlinear energy finite even though the vacuum and kink backgrounds are not in $L^2(\mathbb{R})$?

Answer. The energy uses $b' + v_x$, v_t and $1 - (b+v)^2$. The last quantity is $(1 - b^2) - 2bv - v^2$, whose terms are in L^2 by boundedness of b and the one-dimensional embedding for v.

Self-check. Why does spatial smoothing not give an exact solution of the original nonlinear equation? Which error must vanish in the energy argument?

Answer. Convolution does not commute with the cubic force. The smoothed equation has residual $R_\varepsilon = V'(\phi_\varepsilon) - (V'(\phi))_\varepsilon$. Its L^2 norm tends to zero uniformly on compact time intervals, so its pairing with the uniformly bounded smoothed velocity tends to zero.

Self-check. Why does the estimate for $\|v_x\|_2$ contain $\|b'\|_2$ for a kink, while the corresponding vacuum estimate does not?

Answer. Energy bounds ϕ_x, and $v_x = \phi_x - b'$. The triangle inequality adds $\|b'\|_2$. A vacuum has $b' = 0$. The kink has a nonzero spatial derivative even when its perturbation vanishes.

Self-check. Does the zero conserved energy of the focusing periodic example force its solution to vanish? Which step of the global proof fails there?

Answer. No. The nonzero kinetic term is cancelled by the negative potential. Energy is no longer a sum of nonnegative terms, so it does not control $\|u_t\|_2$ or the existence norm. The explicit solution blows up at time T.